Entanglement cost of antisymmetric Werner states

Abstract

For the antisymmetric Werner state α_d, we prove $E_F(\alpha_d^{\otimes n}) = n$ for every $d \geq 2$ and $n \geq 1$, and hence entanglement cost $E_C(\alpha_d) = 1$ ebit. The proof establishes a sharp companion-channel purity inequality for the antisymmetric-subspace channel, together with its equality cases.

We prove that the antisymmetric state α_d has entanglement cost $E_C(\alpha_d) = 1$ ebit for every $d \geq 2$. Yura established the case $d = 3$ [2]; the case $d = 2$ is a pure singlet. Christandl, Schuch, and Winter proved the dimension-independent lower bound $\log_2(4/3)$ and left open whether the cost equals one [3, 4]. The result below establishes that value also for $d \geq 4$, through the exact identity $E_F(\alpha_d^{\otimes n}) = n$. The associated antisymmetric-subspace channel also occurs in the work of Grudka, Horodecki, and Pankowski on minimum-output Rényi entropy [5]. Here the channel estimate is a sharp purity inequality, including its equality cases, from which the entanglement-cost statement follows.

1 Definitions and statements

All Hilbert spaces are finite-dimensional complex Hilbert spaces. Traces and Schatten norms are unnormalized. Let $H_d = \mathbb{C}^d$, $d \geq 2$, and let $X_d = \Lambda^2 H_d$, with $D = \binom{d}{2}$. Its orthonormal coordinate basis is e_{ab}, $a < b$. The canonical isometry and its partial-trace channel are

$$V_d e_{ab} = \frac{|a,b\rangle - |b,a\rangle}{\sqrt{2}}, \qquad \Phi_d(T) = \mathrm{Tr}_B(V_d T V_d^*). \tag{1.1}$$

For a quantum channel $N : \mathcal{L}(Y) \to \mathcal{L}(B)$, with nonzero input and output spaces, define

$$P_{\max}(N) = \max_{\|y\|=1} \mathrm{Tr}\, N(|y\rangle\langle y|)^2. \tag{1.2}$$

Compactness and continuity give an attained maximum. If $m = \dim B$, every output density has purity at least $1/m$, so $P_{\max}(N) > 0$. A normalized Slater is an element $u = x \wedge y \in X_d$ with $x, y \in H_d$ orthonormal. Thus its edge coordinates are $u_{ab} = x_a y_b - x_b y_a$, and its physical vector is $(x \otimes y - y \otimes x)/\sqrt{2}$. An overall unit phase can be absorbed into either factor.

Theorem 1.1 (Sharp companion bound and equality cases)**.** *Let $d \geq 2$, let N be any channel on its entire input space Y, and set $\beta = P_{\max}(N)$. For every unit $\eta \in X_d \otimes Y$, put $\sigma = \mathrm{Tr}_{X_d}|\eta\rangle\langle\eta|$ and $s = \mathrm{Tr}\,\sigma^2$. Then*

$$\mathrm{Tr}[(\Phi_d \otimes N)(|\eta\rangle\langle\eta|)]^2 \leq \frac{\beta}{2} - \frac{\beta}{d+1}(1 - s). \tag{1.3}$$

Moreover,

$$P_{\max}(\Phi_d \otimes N) = \frac{1}{2} P_{\max}(N). \tag{1.4}$$

A unit input attains this maximum if and only if it is $u \otimes y$, where u is a normalized Slater and y is a unit pure-input maximizer of N. For each $d \geq 3$, the coefficient $\beta/(d+1)$ in (1.3) is the largest possible coefficient valid for all channels and all unit inputs with that normalization.

Let F_d exchange the two physical factors and let

$$\alpha_d = \frac{V_d V_d^*}{D} = \frac{I - F_d}{d(d-1)}. \tag{1.5}$$

The entanglement of formation is measured in bits:

$$E_F(\rho) = \inf_{\rho=\sum_j p_j|\psi_j\rangle\langle\psi_j|} \sum_j p_j S_2(\mathrm{Tr}_B|\psi_j\rangle\langle\psi_j|), \qquad S_2(\tau) = -\operatorname{Tr}\tau\log_2\tau. \tag{1.6}$$

The infimum ranges over every finite normalized complex pure ensemble; zero eigenvalues contribute zero to entropy. Tensor factors are regrouped across the full Alice–Bob cut.

Theorem 1.2 (Tensor powers and formation)**.** *For every $d \geq 2$ and integer $n \geq 0$,*

$$P_{\max}(\Phi_d^{\otimes n}) = 2^{-n}, \qquad E_F(\alpha_d^{\otimes n}) = n. \tag{1.7}$$

For $n \geq 1$, maximum-purity inputs are exactly tensor products of normalized Slaters. Equivalently, every unit physical vector supported on $(V_d X_d)^{\otimes n}$ has Alice purity at most 2^{-n}, with equality exactly for physical tensor products of normalized Slaters. At $n = 0$ the space is $\mathbb{C}$, and the maximizing inputs are unit scalars. Every supported physical pure vector has Alice entropy at least n bits.

Corollary 1.3 (Ordinary entanglement cost)**.** *The ordinary vanishing-error LOCC entanglement cost of α_d is one singlet per copy. Every achievable singlet rate is at least one, and rate one is achievable.*

2 The physical replica operator

Write F for the exchange on $X_d \otimes X_d$. The four-particle alternating subspace embeds in $\mathrm{Sym}^2 X_d$; write W for its orthogonal projector, and put

$$Z = \frac{I+F}{2} - W, \qquad J = \frac{I-F}{2}. \tag{2.1}$$

These are mutually orthogonal projectors with sum I. For each four-subset $Q = \{a < b < c < f\}$, define the real symmetric edge matrix B_Q by the complementary-edge entries

$$(ab, cf) : 1, \qquad (ac, bf) : -1, \qquad (af, bc) : 1,$$

including their transposes, and with every other entry zero. If P_Q projects onto the six edges of Q, then $B_Q^2 = P_Q$ and $\operatorname{rank} B_Q = 6$. Set $\Omega_Q = \operatorname{vec} B_Q$, where vectorization uses the edge basis. The vectors $\Omega_Q/\sqrt{6}$ are an orthonormal basis of the embedded four-particle alternating subspace. In particular,

$$\|\Omega_Q\|^2 = 6, \qquad \sum_Q |\Omega_Q\rangle\langle\Omega_Q| = 6W. \tag{2.2}$$

Lemma 2.1 (Replica identity)**.** *The purity kernel of Φ_d is*

$$T_d = \frac{I + F - 6W}{4} = \frac{Z}{2} - W. \tag{2.3}$$

Proof. On four physical factors let $P = P_{\mathrm{a},12}P_{\mathrm{a},34}$ be the projection onto the antisymmetric subspace of each pair, and let A_4 be the full antisymmetrizer. Decomposing its permutation sum into the six cosets of the within-pair permutation subgroup gives

$$6A_4 = P(I + F - F_{13} - F_{14} - F_{23} - F_{24})P.$$

Here F exchanges the two pairs. The two within-pair swaps act as -1 on the range of P. Conjugating by them equates the four compressed cross swaps. The compression of F_{13} is exactly the swap-trick purity kernel T_d. Consequently $6W = I + F - 4T_d$ after transport by $V_d \otimes V_d$. Expanding a normalized four-particle wedge gives $\Omega_Q/\sqrt{6}$ and proves (2.2). This proves the identity on the whole replica space, including the sector J. □

Choose a Stinespring isometry for N and let N^c be its complementary channel. Its Hermitian swap kernel K_N is characterized by

$$\langle u \otimes v | K_N | x \otimes y \rangle = \operatorname{Tr} N(|x\rangle\langle u|) N(|y\rangle\langle v|). \tag{2.4}$$

This kernel need not be positive. The transition identity for a Stinespring isometry gives $K_{N^c} = K_N F_Y$, where F_Y is the companion replica exchange. For example, if the isometry sends x to a coefficient matrix C_x on $B \otimes E$, then

$$N(|x\rangle\langle u|) = C_x C_u^*, \qquad N^c(|x\rangle\langle u|) = (C_u^* C_x)^T;$$

cyclicity of trace proves the kernel identity.

Write the given input as $\eta = \sum_e e \otimes a_e$ and let A be the matrix with columns a_e. Thus $\sigma = AA^*$ and $\|A\|_{\mathrm{HS}}^2 = 1$. Regroup $\Xi = \eta \otimes \eta$ as $(X_d \otimes X_d) \otimes (Y \otimes Y)$. Define

$$\begin{aligned} q &= \operatorname{Tr} N(\sigma)^2, & q^c &= \operatorname{Tr} N^c(\sigma)^2, \\ z &= \langle \Xi, (Z \otimes K_N)\Xi \rangle, & w &= \langle \Xi, (W \otimes K_N)\Xi \rangle, \\ z_0 &= \langle \Xi, (Z \otimes I)\Xi \rangle, & w_0 &= \langle \Xi, (W \otimes I)\Xi \rangle, \\ M_Q &= AB_Q A^T, & C &= \sum_Q \langle \operatorname{vec} M_Q, K_N \operatorname{vec} M_Q \rangle. \end{aligned} \tag{2.5}$$

Every one of these contractions uses the same matrix A.

Lemma 2.2 (Exact scalar identities)**.** *With the definitions above, the actual output purity P satisfies*

$$z_0 + w_0 = \frac{1+s}{2}, \qquad q + q^c = 2(z+w), \qquad C = 6w, \qquad P = \frac{q + q^c - C}{4} = \frac{z}{2} - w. \tag{2.6}$$

Proof. The expectation of the first-site exchange on Ξ is s, by the swap trick and equality of the two marginal spectra of a pure state. This proves the first identity. The identity part of $I + F$ gives q. Since $(F \otimes F_Y)\Xi = \Xi$, its first-site exchange term gives q^c by $K_{N^c} = K_N F_Y$. Thus the symmetric sector gives $q + q^c = 2(z + w)$. Contraction with Ω_Q gives exactly $\operatorname{vec}(AB_Q A^T)$, proving $C = 6w$ from (2.2). Finally (2.3) and the swap trick prove the purity formula. □

3 A rank-six channel contraction estimate

Lemma 3.1 (Coherence bound)**.** *For orthogonal unit companion inputs ψ_j, ψ_l, set*

$$P_j = \operatorname{Tr} N(|\psi_j\rangle\langle\psi_j|)^2, \qquad B_{jl} = \operatorname{Tr} N(|\psi_l\rangle\langle\psi_j|)^2.$$

Then

$$|B_{jl}| \le \min\left\{ \frac{\beta}{2}, \sqrt{P_j P_l} \right\}. \tag{3.1}$$

Proof. Put $O = \operatorname{Tr} N(|\psi_j\rangle\langle\psi_j|)N(|\psi_l\rangle\langle\psi_l|)$ and $X = \operatorname{Tr} N^c(|\psi_j\rangle\langle\psi_j|)N^c(|\psi_l\rangle\langle\psi_l|)$. The two transition Hilbert–Schmidt norms are X and O, respectively, and their squared traces agree. Therefore $|B_{jl}| \le \min(O, X)$. Hilbert–Schmidt Cauchy–Schwarz gives $O \le \sqrt{P_j P_l}$. Average the output purities of $(\psi_j + e^{i\theta}\psi_l)/\sqrt{2}$ and $(\psi_j - e^{i\theta}\psi_l)/\sqrt{2}$. Choosing θ so that the squared transition contributes positively gives

$$4\beta \ge P_j + P_l + 2O + 2X + 2|B_{jl}| \ge 8|B_{jl}|.$$

For the last inequality use $P_j + P_l \ge 2\sqrt{P_j P_l} \ge 2|B_{jl}|$ and $O, X \ge |B_{jl}|$. This proves both bounds. □

Lemma 3.2 (Rank-six estimate)**.** *For every complex symmetric matrix M of rank at most six, define $C_N(M) = \langle \operatorname{vec} M, K_N \operatorname{vec} M\rangle$. Then*

$$C_N(M) \ge -\frac{4\beta}{5}\|\Lambda^2 M\|_1, \qquad C_N(M) \ge -2\beta\|M\|_{\mathrm{HS}}^2. \tag{3.2}$$

Proof. Take a Takagi decomposition $M = \sum_{j=1}^{r} \lambda_j \psi_j \psi_j^T$, where $r \le 6$, $\lambda_j \ge 0$, and the ψ_j are orthonormal. By (2.4),

$$C_N(M) = \sum_j \lambda_j^2 P_j + 2\sum_{j<l} \lambda_j \lambda_l \operatorname{Re} B_{jl}.$$

The weighted average of the two bounds in (3.1) gives $|B_{jl}| \le (4/5)(\beta/2) + (1/5)\sqrt{P_j P_l}$. Set $x_j = \lambda_j \sqrt{P_j}$. Retaining the diagonal terms yields

$$\begin{aligned} C_N(M) &\ge \frac{6\sum_j x_j^2 - (\sum_j x_j)^2}{5} - \frac{4\beta}{5}\sum_{j<l}\lambda_j\lambda_l \\ &\ge -\frac{4\beta}{5}\sum_{j<l}\lambda_j\lambda_l. \end{aligned}$$

The omitted bracket is nonnegative by $r \le 6$. The singular values of $\Lambda^2 M$ are the pair products $\lambda_j\lambda_l$, proving the first inequality. Also $\sum_{j<l}\lambda_j\lambda_l \le (5/2)\sum_j \lambda_j^2$, proving the second. Zero Takagi values require no separate assumption. □

Since $\operatorname{rank}(AB_Q A^T) \le 6$, applying the second estimate to the actual correction matrices gives

$$w \ge -2\beta w_0. \tag{3.3}$$

Indeed, $\sum_Q \|AB_Q A^T\|_{\mathrm{HS}}^2 = 6w_0$ by (2.2).

4 The Slater moment and the positive sector

Lemma 4.1 (Normalized Slater-square moment)**.** *Let $R(U) = \Lambda^2 U$ for $U \in U(d)$ and let $u(U) = R(U)e_{12}$. With normalized Haar measure,*

$$\int_{U(d)} |u(U)\otimes u(U)\rangle\langle u(U)\otimes u(U)|\, dU = \frac{Z}{z_d}, \qquad z_d = \operatorname{rank} Z = \frac{d^2(d^2-1)}{12}. \tag{4.1}$$

Proof. Every Slater square is symmetric and has zero four-form component, because $(x\wedge y)\wedge(x\wedge y) = 0$. The integral is therefore supported on Z and commutes with $R(U)\otimes R(U)$. We verify that a commuting operator on Z is scalar.

Diagonal physical phases separate coordinate weights. The weight of $e_{ab} \otimes e_{ab}$ is one-dimensional, and physical permutations equate the scalars on all these squares. Rotating coordinate b into c in such a square and comparing the middle coefficient forces the same scalar on every common-index symmetric pair $e_{ab} \odot e_{ac}$. Here $p \odot q = (p \otimes q + q \otimes p)/\sqrt{2}$ for orthogonal p, q.

For four distinct $a < b < c < f$, let $x = e_{ab} \odot e_{cf}$, $y = e_{ac} \odot e_{bf}$, $t = e_{af} \odot e_{bc}$. The four-form line is $x - y + t$, and its orthogonal complement is the two-dimensional Z weight space. Rotate a into c in $e_{ab} \odot e_{af}$: after removing the common-index terms the middle coefficient is $x - t$. Rotating a into f in $e_{ab} \odot e_{ac}$ similarly produces $-(x + y)$. Commutation gives the same scalar on these two independent vectors. Coordinate squares, common-index pairs, and these weight planes span Z, proving the assertion. The argument also covers $d = 3$, where the four-distinct weight spaces are absent, and $d = 2$ is one-dimensional.

The integral has trace one. Finally

$$z_d = \frac{D(D+1)}{2} - \binom{d}{4} = \frac{d^2(d^2-1)}{12},$$

which proves the stated normalization. □

For a normalized Slater u, contracting the first-site square of the fixed Ξ gives $v_u \otimes v_u$, with $v_u = A\overline{u}$. The definition of β on the whole companion input space gives

$$\langle v_u \otimes v_u, K_N(v_u \otimes v_u)\rangle = \operatorname{Tr} N(|v_u\rangle\langle v_u|)^2 \le \beta \|v_u\|^4.$$

This is homogeneous and also holds when $v_u = 0$. Integrating and using (4.1) on both sides yields

$$z \le \beta z_0. \tag{4.2}$$

Define $\Delta = \beta(1+s) - (q + q^c)$. The exact identities (2.6) now give

$$C + 2\Delta = 4(\beta z_0 - z) + 2(w + 2\beta w_0) \ge 0. \tag{4.3}$$

Thus $C + \Delta \ge C/2$. This conclusion uses contractions on the given $\eta \otimes \eta$; it does not require K_N to be positive.

5 The invariant exterior correction bound

Lemma 5.1 (Common-matrix exterior bound)**.** *For $d \ge 4$, the same correction C satisfies*

$$C \ge -2\beta \frac{d-3}{d+1}(1-s). \tag{5.1}$$

Proof. For a physical unitary U, write $R = \Lambda^2 U$. The vectors $(R \otimes R)\Omega_Q/\sqrt{6}$ form another orthonormal basis of the same four-form subspace. Their actual bra contractions are

$$M'_Q = A\overline{RB_QR^T}A^T.$$

Accordingly $C = \sum_Q C_N(M'_Q)$ for every U. Each matrix still has rank at most six. Put $L = \Lambda^2 A$. The symmetric matrix $\Lambda^2\overline{RB_QR^T}$ is a partial isometry of rank fifteen; its left support is

$$Q'_Q = \Lambda^2(\overline{R}P_QR^T).$$

All its nonzero Takagi values are one, since the six nonzero singular values of B_Q are one. Hence it has a factorization VV^T with $V^*V = I_{15}$ and $VV^* = Q'_Q$. Multiplicativity of exterior powers and Schatten Cauchy–Schwarz imply

$$\|\Lambda^2 M'_Q\|_1 = \|(LV)(LV)^T\|_1 \le \|LV\|_{\mathrm{HS}}^2 = \operatorname{Tr} L^*LQ'_Q. \tag{5.2}$$

The first estimate of Lemma 3.2 therefore gives, for every U,

$$C \ge -\frac{4\beta}{5} \operatorname{Tr} L^*L \sum_Q Q'_Q. \tag{5.3}$$

The unitary action on $\Lambda^2(\Lambda^2 H_d)$ has scalar commutant. Here is a coordinate verification. Its one-dimensional repeated-index weight spaces are spanned by $e_{ab} \wedge e_{ac}$. Permutations equate their scalars. For four distinct $a < b < c < f$, set $x = e_{ab} \wedge e_{cf}$, $y = e_{ac} \wedge e_{bf}$, $t = e_{af} \wedge e_{bc}$. Rotating the common index a into f, c, and b in, respectively, $e_{ab} \wedge e_{ac}$, $e_{ab} \wedge e_{af}$, and $e_{ac} \wedge e_{af}$ produces the middle coefficients

$$-x + y, \qquad x + t, \qquad y - t.$$

The remaining coefficients are repeated-index vectors already controlled. These three middle coefficients are independent and span the four-distinct weight space. Commutation forces the same scalar on it. All coordinate outer wedges belong to one of these types, proving the assertion. Conjugating the representation preserves this property.

Consequently the Haar average of $\sum_Q Q'_Q$ is scalar. Its trace is $15\binom{d}{4}$, and the ambient dimension is $\binom{D}{2}$, so

$$\int \sum_Q Q'_Q \, dU = R_d I, \qquad R_d = \frac{15\binom{d}{4}}{\binom{D}{2}} = \frac{5(d-3)}{d+1}. \tag{5.4}$$

The matrix L stays fixed during this integration. If a_j are the singular values of A, then $\sum_j a_j^2 = 1$ and

$$\|L\|_{\mathrm{HS}}^2 = \sum_{j<l} a_j^2 a_l^2 = \frac{1 - \operatorname{Tr}(AA^*)^2}{2} = \frac{1-s}{2}.$$

Integrating (5.3) proves (5.1). □

6 Sharpness and rigidity

Proof of Theorem 1.1. For $d \ge 4$, combine (2.6), (4.3), and (5.1):

$$\begin{aligned} P &= \frac{\beta(1+s) - (C+\Delta)}{4} \\ &\le \frac{\beta(1+s) + \beta\frac{d-3}{d+1}(1-s)}{4} = \frac{\beta}{2} - \frac{\beta}{d+1}(1-s). \end{aligned}$$

For $d = 3$, $W = 0$ and (4.2) gives $P \le (\beta/4)(1+s)$, the same formula. For $d = 2$, X_d is one-dimensional, so every input is a product and $s = 1$; its first output has purity $1/2$.

For a unit $u \in X_d$, Lemma 2.1 gives directly

$$\operatorname{Tr} \Phi_d(|u\rangle\langle u|)^2 = \frac{1}{2} - \frac{3}{2}\|W(u \otimes u)\|^2 \le \frac{1}{2}. \tag{6.1}$$

A coordinate Slater attains $1/2$. Thus a product with any pure-input maximizer of N attains $\beta/2$, proving (1.4).

For the complete equality case of (6.1), let u_{ab} be extended as a skew matrix to all ordered coordinates. The condition $W(u \otimes u) = 0$ is exactly the family of Plücker relations

$$u_{ab}u_{cf} - u_{ac}u_{bf} + u_{af}u_{bc} = 0 \quad \text{for all } a, b, c, f. \tag{6.2}$$

Indeed these are the coefficients of $u \wedge u$; the normalized four-form basis above has nonzero constant normalization. Choose a pivot $u_{ab} \neq 0$. Applying (6.2) to arbitrary i, j yields

$$u_{ij} = \frac{u_{ai}u_{bj} - u_{aj}u_{bi}}{u_{ab}}.$$

Thus $u = x \wedge y$ with $x_i = u_{ai}$ and $y_i = u_{bi}/u_{ab}$. Gram–Schmidt, together with $\|x \wedge y\|^2 = \|x\|^2\|y\|^2 - |\langle x, y\rangle|^2$, writes the same unit u as a wedge of orthonormal vectors, with its phase absorbed in one factor. Conversely every such wedge satisfies (6.2). Therefore one-copy equality holds exactly for normalized Slaters.

Since $\beta/(d+1) > 0$, equality at the maximum in (1.3) forces $s = 1$. A pure bipartite vector with a pure marginal is a product, say $\eta = u \otimes y$ with both factors unit. Its output purity is the product of the two local output purities. Equality at $\beta/2$, the positive product of their upper bounds, forces local purity $1/2$ and companion purity β. The converse is immediate.

To prove coefficient sharpness for $d \geq 3$, use $N = \Phi_d$ and

$$\eta_{\text{flat}} = D^{-1/2} \sum_e e \otimes e. \tag{6.3}$$

Then $s = 1/D$ and $\beta = 1/2$. Its physical coefficient matrix after regrouping outputs (a, c) and environments (b, f) is

$$\frac{1}{2\sqrt{D}}(\delta_{ac}\delta_{bf} - \delta_{af}\delta_{bc}).$$

Tracing the environment gives the literal output density

$$\tau_{\text{flat}} = \frac{I + (d-2)|v\rangle\langle v|}{4D}, \qquad v = \sum_{a=1}^{d} |a, a\rangle, \quad \|v\|^2 = d. \tag{6.4}$$

Its eigenvalues are $(d-1)^2/(4D)$ on the v line and $1/(4D)$ with multiplicity $d^2 - 1$. Consequently

$$\operatorname{Tr} \tau_{\text{flat}}^2 = \frac{d^2 - 3d + 4}{4d(d-1)} = \frac{1}{4} - \frac{1 - 1/D}{2(d+1)}. \tag{6.5}$$

This is equality in (1.3). As $D > 1$ for $d \geq 3$, increasing its impurity coefficient would fail at this input. At $d = 2$ every possible input has zero impurity. □

7 Tensor powers, entropy, and operational cost

Proof of Theorem 1.2. The zero-copy channel on $\mathbb{C}$ has maximum purity one. Theorem 1.1 applied successively with $N = \Phi_d^{\otimes(n-1)}$ proves the first identity in (1.7). Its equality characterization also inducts: a maximizing input factors as a first normalized Slater times a maximizing input on the remaining sites. The base case is a unit scalar, so all phases are included. Conversely every Slater tensor product attains 2^{-n}.

The isometry $V_d^{\otimes n}$ identifies every unit vector of $X_d^{\otimes n}$ with a unit physical vector in the antisymmetric support. Its Alice reduction is exactly the output of $\Phi_d^{\otimes n}$. This proves both physical purity assertions. For a density with eigenvalues λ_j, concavity of logarithm with weights λ_j gives

$$S_2(\tau) = -\sum_{\lambda_j>0} \lambda_j \log_2 \lambda_j \geq -\log_2 \sum_j \lambda_j^2.$$

The purity bound therefore gives entropy at least n on every supported pure vector.

Let P_n project onto the physical antisymmetric support. If $\alpha_d^{\otimes n} = \sum_j p_j |\psi_j\rangle\langle\psi_j|$, then

$$0 = \operatorname{Tr}(I - P_n)\alpha_d^{\otimes n} = \sum_j p_j \|(I - P_n)\psi_j\|^2.$$

Every positive-weight component lies in the support and hence has entropy at least n. This proves $E_F \geq n$ for every unrestricted finite pure ensemble. The uniform ensemble of all D^n coordinate Slater tensor products has barycenter $\alpha_d^{\otimes n}$. Each component has n independent two-point Schmidt spectra and therefore entropy n. It proves $E_F \leq n$. The empty tensor product has entropy zero, covering $n = 0$. □

For completeness, the operational quantity in Corollary 1.3 uses shared two-qubit singlets, free local ancillas, local quantum instruments, and classical communication. Convergent output limits of finite protocol prefixes are allowed. An HHT achievable rate r means that for every $\epsilon, \delta > 0$ there are integers $n \geq 1, m \geq 0$ and such a protocol with output τ for which

$$\left| r - \frac{m}{n} \right| \leq \delta, \qquad D_{\mathrm{B}}(\tau, \alpha_d^{\otimes n}) \leq \epsilon. \tag{7.1}$$

Here $D_{\mathrm{B}}(\rho, \sigma) = 2\sqrt{1 - \operatorname{Tr}\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}}}$. The cost is the infimum of achievable rates. The same rates result from finite protocols: a convergent output can be approximated by a finite prefix on the same charged input, changing only the output error; a finite protocol is an eventually constant convergent process. Local basis embeddings and appending or discarding free local ancillas also preserve the rate and final output density.

Proof of Corollary 1.3. The regularization theorem of Hayden, Horodecki, and Terhal [1, Theorem 1] identifies this ordinary LOCC cost with

$$E_C(\rho) = \lim_{n\to\infty} \frac{E_F(\rho^{\otimes n})}{n}. \tag{7.2}$$

Its operational approximation and metric conventions are those of (7.1); equivalently one may use vanishing trace-distance error. The theorem is used here as the established operational bridge. By Theorem 1.2 the right side for α_d is one. Thus every achievable rate is at least one.

There is also an explicit rate-one finite protocol. Alice selects an edge $\{a, b\}$ uniformly, sends its classical label to Bob, and both apply the local isometry $|0\rangle \mapsto |a\rangle$, $|1\rangle \mapsto |b\rangle$ to one singlet. The resulting physical state is $V_d e_{ab}$. Forgetting the label gives α_d. Repeating independently creates $\alpha_d^{\otimes n}$ exactly from n singlets, proving that rate one itself is achievable. □

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